

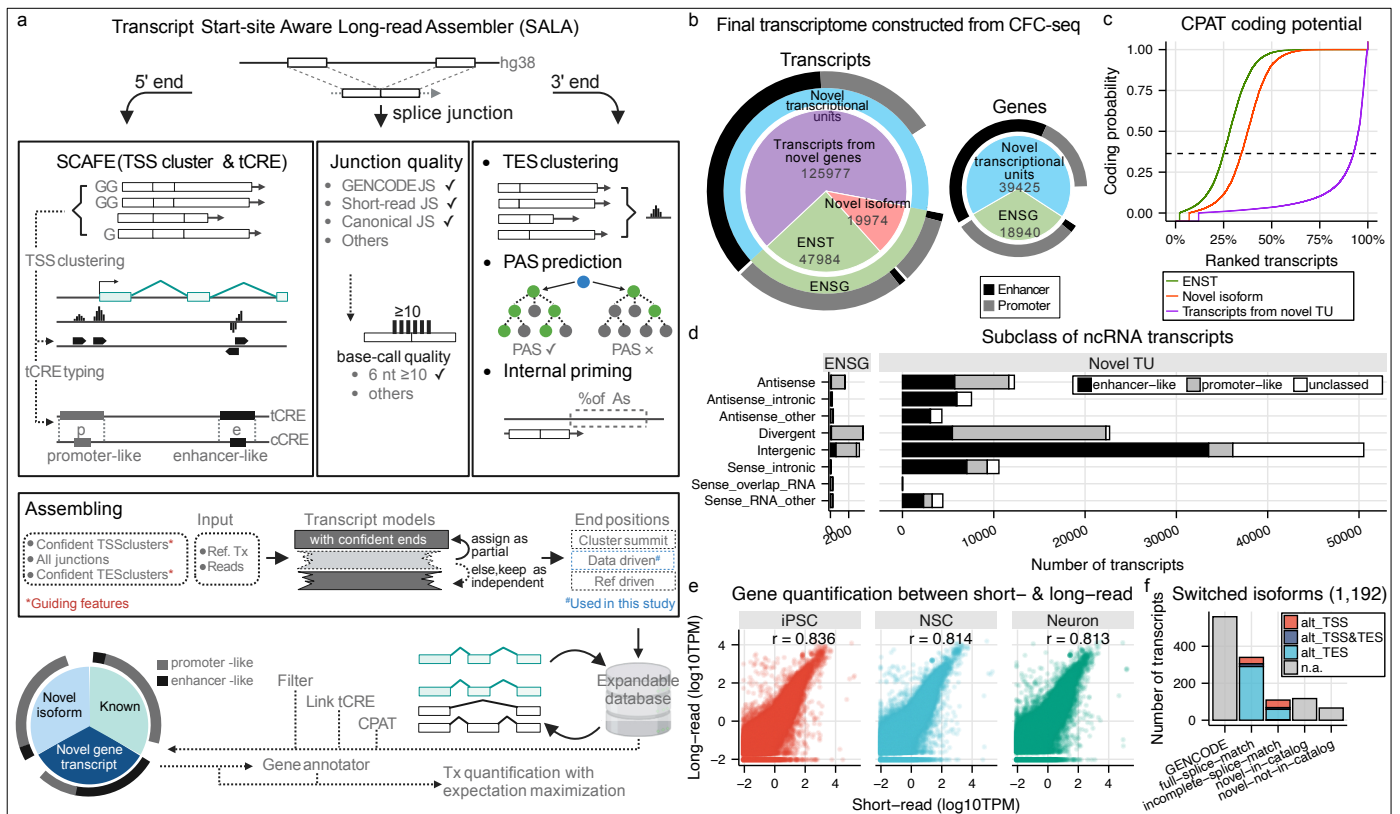


Figure 3 | Transcript models identified by SALA.

a, Overview of the analytical pipeline. Three transcript features—5' end, splice junctions, and 3' end—were extracted from all reads. TSS clusters were defined and filtered by SCAFE; splice junctions were evaluated against GENCODE (v39), matched short-read RNA-seq, and base-calling quality summarization; TES clusters were formed and filtered by read count, with additional 3' end labels from PAS and internal priming detection. These features, along with all reads and the reference transcriptome, were used for guided transcript assembly. The resulting transcript models were annotated with tCRE-based promoter types and filtered for downstream analysis. Final quantification was performed using expectation maximization in Bambu. **b**, Number of transcript and gene models identified by SALA. Transcript models were classified into GENCODE-annotated (ENST), novel isoforms (within known genes), and transcripts from novel transcriptional units. **c**, Coding potential of transcript groups assessed using CPAT. **d**, All detected ncRNAs, including annotated lncRNAs and novel ncRNAs, were subclassified based on their location relative to protein-coding genes and pseudogenes. **e**, Comparison of gene-level expression from long- and short-read quantification using the finalized transcriptome and detectable GENCODE models, ribosomal RNAs were excluded. **f**, Classes of transcript models that exhibit significant isoform switching across iPSC, NSC, and Neuron, only isoforms from GENCODE genes were shown. Alternative (alt) TSS and TES represent novel transcript models with unannotated TSS and TES respectively.